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ABSTRACT

The project titled “Calculating Family Expenses using ServiceNow” focuses on developing an automated expense management system using the ServiceNow Platform-as-a-Service (PaaS) environment. The main objective of this project is to simplify the process of recording, organizing, and analyzing daily family expenditures in a centralized, cloud-based application.

Traditional methods of tracking expenses—such as manual entries or spreadsheets—often lead to data inconsistency, human errors, and lack of analytical insights. This project leverages ServiceNow’s workflow automation, form design, and report generation features to create a dynamic application capable of managing multiple expense categories such as food, travel, utilities, and education.

The system allows users to input their expenses through customized catalog forms, which are stored in a relational table structure. Automated workflows calculate the total monthly expenditure and generate summary reports that can be viewed in real-time dashboards. Client scripts and business rules are implemented to validate user inputs and ensure accuracy of calculations.

By using Flow Designer and Report Builder in ServiceNow, the project achieves seamless automation without extensive coding. The developed system can be further extended to include predictive analytics and mobile access through the ServiceNow Mobile App.

Overall, this project demonstrates how the ServiceNow platform can be utilized beyond IT Service Management (ITSM) to develop custom business applications that improve efficiency,

transparency, and data accuracy in everyday life scenarios such as family expense management.

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CHAPTER 1

INTRODUCTION

1.1 Overview

In the modern digital era, managing household finances has become an essential part of maintaining economic stability. Families often find it difficult to manually record and monitor their daily expenses due to busy lifestyles and lack of systematic tracking. Traditional methods such as notebooks and spreadsheets are time-consuming, error-prone, and lack automation.

The project titled “Calculating Family Expenses using ServiceNow” is designed to provide an automated, efficient, and user-friendly platform for recording, calculating, and analyzing family expenditures. ServiceNow, a cloud-based development platform, is utilized for building this custom application using its low-code environment, workflow automation, and reporting capabilities.

1.2 Need for Automation

Manual expense tracking often leads to data inconsistency, missing records, and poor financial analysis. Automation not only saves time but also improves data accuracy and provides meaningful insights into spending patterns. By implementing this system in ServiceNow, the user can access expense data securely from anywhere through the cloud.

1.3 Objectives of the Project

- To design and develop an automated expense calculation system using ServiceNow.
- To categorize expenses such as food, travel, education, and bills.

- To calculate total monthly and category-wise expenditures automatically.
- To display interactive dashboards with graphical reports.
- To improve data security and reduce manual calculation errors.

1.4 Scope of the Project

The project is limited to developing a family expense management application within the ServiceNow platform. The application includes modules for user login, expense entry, and automated calculation. It provides basic reporting and dashboard visualization using ServiceNow's inbuilt tools.

1.5 Significance

This project demonstrates how ServiceNow can be applied beyond IT Service Management (ITSM) into real-world data management solutions. It helps students understand platform capabilities such as form creation, workflow automation, scripting, and reporting — valuable skills for cloud and enterprise development.

CHAPTER 2

LITERATURE SURVEY

2.1 Existing Systems

Common expense tracking systems such as Excel-based sheets, Google Sheets, and standalone Android apps require manual data entry and lack real-time synchronization across devices. These systems also do not support workflow automation or centralized storage.

2.2 Limitations of Manual Methods

- High probability of human error.
- Lack of visualization and reports.
- No reminder or alert system.
- Difficulty in analyzing monthly trends.

2.3 Cloud-Based Solutions

Cloud-based platforms such as ServiceNow and Salesforce allow rapid application development with minimal coding. They offer automation, security, and integration features. Using ServiceNow, applications can be built using tables, forms, flows, and scripts, making it ideal for this project.

2.4 Summary

After reviewing existing systems, ServiceNow was chosen as the platform due to its no-code/low-code capabilities, scalability, and reporting tools, which make it suitable for automating family expense management efficiently.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Problem Definition

The main problem addressed in this project is the lack of an automated system for tracking and analyzing household expenses efficiently. The proposed system eliminates manual entry errors and provides accurate real-time summaries of family expenses.

3.2 Feasibility Study

- **Technical Feasibility:** ServiceNow provides a cloud-based development environment that supports table creation, workflow management, and data visualization.
- **Operational Feasibility:** The system can be easily accessed via a web browser and requires minimal training.
- **Economic Feasibility:** As the project uses the Naan Mudhalvan academic ServiceNow environment, implementation cost is negligible.

3.3 Requirements Analysis

Hardware Requirements:

- Laptop or desktop computer
- Internet connection

Software Requirements:

- ServiceNow Developer Instance
- Web Browser (Chrome or Edge)

3.4 Functional Requirements

- Add new expense records
- Categorize expenses (Food, Travel, Bills, etc.)
- Calculate monthly and category-wise totals
- Generate reports and dashboards

3.5 Non-Functional Requirements

- High data accuracy
- Secure access through ServiceNow authentication
- Fast data retrieval and dashboard loading

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The architecture of the system follows a client-server model, where users input expense data through forms, which are stored in ServiceNow tables. Business Rules and Flows handle the calculations and report generation.

User → ServiceNow Form → Business Rule/Flow → Database Table
→ Report/Dashboard

4.2 Data Flow Diagram

1. Level 0 (Context Diagram): Shows user submitting data to the system.
2. Level 1: Data stored in tables like family_expense and expense_category.
3. Level 2: Flow Designer triggers to calculate total monthly expenses.

4.3 UML Diagrams

- Use Case Diagram: User can Add, Edit, and View Expenses.
- Activity Diagram: Flow from expense entry to dashboard visualization.
- Class Diagram: Tables and their relationships (User, Expense, Category).

4.4 Table Design

Table Name	Description
family_expense	Stores expense details like date, category, and amount.
expense_category	Defines categories like Food, Travel, Rent, etc.
user_profile	Stores user details and login information.

4.5 Form Design

Each form in ServiceNow includes fields like Date, Expense Type, Description, Amount, and Payment Mode. Form layout is designed using Form Designer and enhanced using Client Scripts for validation.

CHAPTER 5

MODULE DESCRIPTION

5.1 User Module

Handles user login, profile management, and authentication using ServiceNow's Access Control Lists (ACLs).

5.2 Expense Entry Module

Allows users to record daily expenses under various categories. Validation ensures correct data entry and prevents duplicates.

5.3 Calculation Module

Automatically calculates total daily and monthly expenses using Business Rules or Flow Designer logic.

5.4 Dashboard Module

Displays graphical reports and summary tables. ServiceNow's Report Builder is used to generate pie charts and bar graphs showing expenses per category.

5.5 Admin Module

Admin users can add or remove categories, manage data records, and view consolidated reports for all users.

CHAPTER 6

IMPLEMENTATION

6.1 Tools and Technologies Used

The system is implemented using ServiceNow, a cloud-based platform that allows developers to create and deploy custom applications with minimal coding effort. The following tools and technologies were used:

Tool/Technology	Purpose
ServiceNow Studio	Application design and development
Flow Designer	Workflow automation
Business Rules	Logic-based automation
Client Scripts	Front-end validation
Tables and Forms	Data storage and entry interface
Reports and Dashboards	Data visualization and analysis

6.2 Creating Tables

The Family Expense Application consists of three main tables:

1. User Profile Table: Stores user credentials and details.
2. Expense Category Table: Maintains various expense types such as Food, Travel, and Utilities.
3. Family Expense Table: Records all expense transactions with date, amount, and category.

6.3 Form and Field Configuration

Forms were designed using Form Designer in ServiceNow. Each form includes fields such as:

- Expense Date
- Expense Type (Dropdown)
- Amount

- Description
- Payment Method

Form validation was added using Client Scripts to ensure that all required fields are filled before submission.

6.4 Business Rules

- Business Rules are used to automatically calculate total expenses when a new record is inserted or updated. For example:
- A rule calculates the monthly total each time a new expense is added.
- Another rule triggers an email notification when the monthly expense exceeds a threshold.

6.5 Flow Designer

Flow Designer automates processes like:

- Adding new expense records.
- Triggering recalculations of totals.
- Updating the dashboard automatically.

These flows are designed graphically without complex scripting.

6.6 Reports and Dashboards

ServiceNow's Report Builder is used to create:

- Bar charts of monthly expenses.
- Pie charts showing category-wise distribution.

- Summary tables comparing weekly and monthly spending.

Dashboards combine multiple reports into a single interface for real-time analysis.

CHAPTER 7

TESTING AND RESULTS

7.1 Testing Methods

Testing ensures that the application works as expected. The following methods were performed:

- Unit Testing: Individual modules such as expense entry and calculation were tested.
- Integration Testing: Interaction between modules (e.g., adding expense updates dashboard).
- User Acceptance Testing: Verified by users for ease of use and data accuracy.

7.2 Sample Test Cases

Test	Description	Expected	Output	Result
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Case ID

TC01	Add new expense. Record stored successfully	Pass
TC02	Calculate total Total updated correctly	Pass
TC03	Invalid amount Show error message	Pass
	input	
TC04	Generate report Report displays data correctly	Pass

7.3 Output Screens (Example)

- Expense form with validation message
- Monthly expense dashboard
- Pie chart of category-wise spending

7.4 Result Analysis

The testing phase confirmed that:

- The system calculates totals accurately.
- The user interface is intuitive and easy to use.
- Data updates reflect immediately in dashboards.
- Automation reduces manual effort significantly.

CHAPTER 8

ADVANTAGES AND LIMITATIONS

8.1 Advantages

- Automated tracking of expenses with minimal user input.
- Cloud-based access — available from anywhere.
- Real-time dashboards for visual insights.
- Data stored securely on the ServiceNow platform.
- Customizable workflows for different family members.

8.2 Limitations

- Requires ServiceNow developer or instance access.
- Limited offline functionality without internet.
- Restricted customization for free-tier accounts.

CHAPTER 9

FUTURE ENHANCEMENTS

The current system can be enhanced further to include:

1. Mobile Application Integration:

Integrating with ServiceNow Mobile Studio to allow users to log expenses from smartphones.

2. AI-based Expense Prediction:

Using predictive analytics to forecast future spending patterns based on past data.

3. Multi-user Family Mode:

Allowing multiple family members to enter expenses into a shared database.

4. Cloud API Integration:

Connecting with Google Sheets or banking APIs for real-time financial updates.

5. Voice Entry Feature:

Adding voice input for quicker expense logging.

These enhancements can improve accessibility and provide deeper financial insights for users.

CHAPTER 10

CONCLUSION

This project, “Calculating Family Expenses using ServiceNow,” successfully demonstrates how the ServiceNow platform can be used beyond its traditional IT service management functions to develop custom cloud-based business applications.

Through this project, a functional expense management system was designed and implemented. It automates the process of adding, calculating, and visualizing family expenditures using ServiceNow’s workflow automation, form design, and reporting modules.

The system achieved its objectives of reducing manual effort, minimizing errors, and providing meaningful visual insights into financial data. Users can now easily view their monthly and category-wise spending through intuitive dashboards.

This project has strengthened the understanding of:

- Platform development using ServiceNow Studio.
- Implementation of Flow Designer and Business Rules.
- Building reports and dashboards for real-time visualization.

In conclusion, this project showcases how cloud-based low-code platforms like ServiceNow can be leveraged by students and professionals to create efficient, scalable, and secure solutions for everyday life challenges such as family expense management.

